

Cahier de Charges (Specifications Document)

Project: KargoConnect

- **Version:** 1.2
- **Date:** August 4, 2025
- **Author:** [Your Name/Company Name]
- **Project Stakeholders:** Project Owner, Development Team

1. Introduction & Project Vision

1.1. Purpose of this Document

This document defines the functional and non-functional requirements for the "KargoConnect" platform, Version 1.2. It serves as the primary source of truth for the development team to understand the project's scope, objectives, and constraints.

1.2. Project Vision

KargoConnect is a peer-to-peer logistics marketplace designed to connect individuals and businesses with unused vehicle space (**Transporters**) to those who need to ship goods (**Senders**). The platform aims to create a more affordable, efficient, and community-driven alternative to traditional shipping services, reducing costs for senders and creating a new revenue stream for transporters.

1.3. Business Objectives

- Provide Senders with a shipping solution that is up to 50% cheaper than traditional options.
- Enable Transporters to monetize their empty vehicle space.
- Build a trusted community through verification, secure payments, and a transparent review system.
- Achieve a critical mass of 1,000 active users (Senders and Transporters) within the first 6 months post-launch.
- Establish an eco-friendly shipping alternative by optimizing existing journeys.

2. Project Scope

2.1. In Scope (V1.2)

- **Fully Responsive Web Application:** A seamless experience on desktop, tablet, and mobile browsers, featuring adaptive layouts for all pages.
- **Role-Based User System:** Multi-step user registration and comprehensive profile management for two primary roles: **Sender** and **Transporter**.
- **Dynamic User Dashboards:**

- Modern, overlapping design with a customizable cover photo.
- Private dashboard view for users to manage their account, trips/demands, and bookings.
- Public profile view for visitors, showcasing reputation, reviews, and completed activities.
- **Core Marketplace Functionality:**
 - Full lifecycle for trip posting by Transporters.
 - Full lifecycle for demand (package) posting by Senders.
- **Advanced Search & Discovery:**
 - A dual-mode search page to find both trips and package demands.
 - AI-powered search assistant using natural language queries (Gemini API).
 - Search results displayed in a responsive table on desktop and a card-based layout on mobile/tablet.
 - Pagination for all lists and search results.
- **Booking & Offer System:**
 - Detailed view pages for both trips and demands.
 - A booking system for Senders with modals for payment (Stripe) or free confirmation.
 - An offer system for Transporters with loading and success confirmation modals.
- **AI-Powered Enhancements (Gemini API):**
 - AI-generated summaries for trips.
 - AI-generated handling instructions for packages.
- **Community & Trust Features:**
 - Real-time chat between users for booked shipments.
 - A two-way rating and review system accessible from public profiles.
- **Internationalization:**
 - Full multi-language support for English, French, and Arabic, including RTL layout for Arabic.
- **Notifications:**
 - Basic in-app notification system for key events.

2.2. Out of Scope (For future versions)

- Native iOS and Android mobile applications.
- Advanced B2B/enterprise features (e.g., fleet management).
- Integrated shipping insurance provider.

- Real-time geospatial tracking of active deliveries.
- Complex route optimization algorithms.

3. Functional Requirements (User Stories & Tasks)

Epic 1: User Account & Profile Management

- **User Story 1.1:** As a new user, I want to choose my role and sign up through a validated, multi-step form, so that I can provide all necessary information easily.
- **Tasks:**
 - [Frontend] Create the multi-step registration form component with `react-hook-form`.
 - [Frontend] Implement a modern, drag-and-drop image uploader for vehicle and profile pictures.
 - [Frontend] Add client-side validation for all required fields, with internationalized error messages.
- **User Story 1.2:** As a user, I want to log in securely, so that my account is protected.
- **Tasks:**
 - [Frontend] Create a modern, responsive login form.
- **User Story 1.3:** As a user, I want a personal dashboard to manage my account and activity, so that I have a central place to control my information.
- **Tasks:**
 - [Frontend] Create a unified dashboard component controlled by an `isOwnerView` prop.
 - [Frontend] Design a modern layout with a changeable cover photo and an overlapping profile card.
 - [Frontend] Implement tabbed navigation with an active-state underline for different sections (Settings, Trips/Demands, etc.).
 - [Frontend] Create responsive data tables that convert to a card view on mobile and tablet devices, including search, pagination, and icon-based action menus.
- **User Story 1.4:** As a visitor, I want to view a user's public profile, so that I can assess their reputation and history before interacting with them.
- **Tasks:**
 - [Frontend] Implement the public view of the dashboard component.

- [Frontend] Display key public information: reviews, completed trips/demands, rating, and verification status.
- [Frontend] Create a "Leave a Review" component on public profiles.

Epic 2: Trip & Demand Management

- **User Story 2.1:** As a Transporter, I want to post and manage my trips from my dashboard, so that I can offer my available space to Senders.
- **Tasks:**
 - [Backend] Create POST /trips and GET /users/me/trips endpoints.
 - [Frontend] Create the "Post a Trip" form/modal.
 - [Frontend] Implement the "My Trips" tab in the transporter dashboard with responsive lists.
- **User Story 2.2:** As a Sender, I want to post and manage my package demands from my dashboard, so that Transporters can make offers.
- **Tasks:**
 - [Backend] Create POST /demands and GET /users/me/demands endpoints.
 - [Frontend] Create the "Post New Demand" form/modal.
 - [Frontend] Implement the "My Demands" tab in the sender dashboard with responsive lists.

Epic 3: Search & Booking

- **User Story 3.1:** As a user, I want to search for both trips and package demands on a single, easy-to-use page, so that I can find what I need efficiently.
- **Tasks:**
 - [Frontend] Create a unified search page with a toggle to switch between "Find a Transporter" and "Find a Package" modes.
 - [Frontend] Implement skeleton loaders during search and pagination.
- **User Story 3.2:** As a Sender, I want to view trip details and book with a clear confirmation process, so that I am confident in my booking.
- **Tasks:**
 - [Frontend] Create the "Trip Details" page with a two-column layout.
 - [Frontend] Implement a multi-step booking modal for payment (with credit card and PayPal options) and free confirmations.
- **User Story 3.3:** As a Transporter, I want to view demand details and make an offer, so that I can bid to transport a package.

- **Tasks:**
 - [Frontend] Create the "Demand Details" page.
 - [Frontend] Implement an offer submission flow with loading and success confirmation modals.

Epic 4: AI-Powered Features

- **User Story 4.1:** As a user, I want to use natural language for my search, so that the system can automatically fill in the form for me.
- **Tasks:**
 - [Frontend] Integrate the Gemini API for natural language understanding on the search page.
- **User Story 4.2:** As a user, I want to get a quick, AI-generated summary of a trip or package, so that I can understand the key details faster.
- **Tasks:**
 - [Frontend] Integrate the Gemini API for text generation on trip/demand detail cards.

Epic 5: Notifications

- **User Story 5.1:** As a user, I want to be notified of important events, so that I can respond in a timely manner.
- **Tasks:**
 - [Backend] Create a notifications service and GET /users/me/notifications endpoint.
 - [Backend] Implement logic to generate notifications for: new offers, accepted/declined offers, new bookings, booking confirmations, and new chat messages.
 - [Frontend] Create a notification icon and dropdown menu in the main site header.

4. Non-Functional Requirements

- **Performance:**
 - Page Load Time: All critical pages must load in under 3 seconds.
 - Delayed Loading UI: Loading indicators should only appear after 300ms to avoid UI flicker on fast connections.
- **Usability:**
 - The user interface must be fully responsive and optimized for desktop, tablet, and mobile devices.
 - The application must support LTR and RTL layouts for internationalization.

- **Reliability & Security:**

- Uptime target of 99.5%.
- All web traffic served over HTTPS.
- Secure password hashing and PCI-compliant payment integration.

5. Technical Constraints

- **Frontend:** Next.js (React), react-hook-form, next-intl, Tailwind CSS.
- **Backend:** NestJS (Node.js)
- **Database:** PostgreSQL with PostGIS extension
- **Authentication:** NextAuth.js
- **AI Features:** Gemini API
- **Real-time Communication:** WebSockets (via NestJS Gateways)

6. Deliverables

- A fully functional, responsive, and internationalized web application meeting all requirements.
- Complete source code managed in a Git repository.
- API documentation for all backend services.